TUGAS PRATIKUM
JARINGAN KOMPUTER 6

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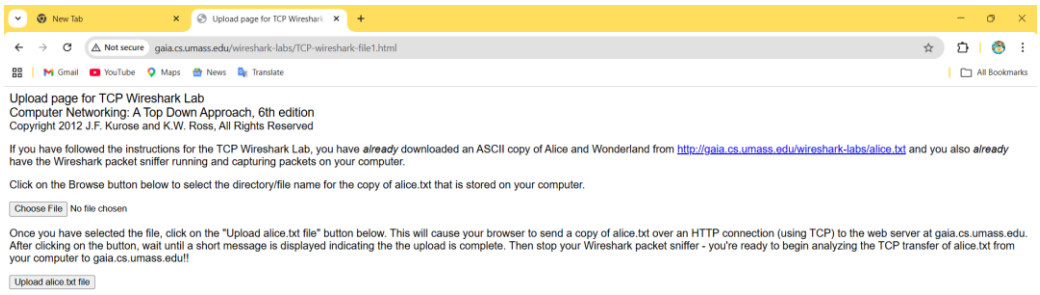
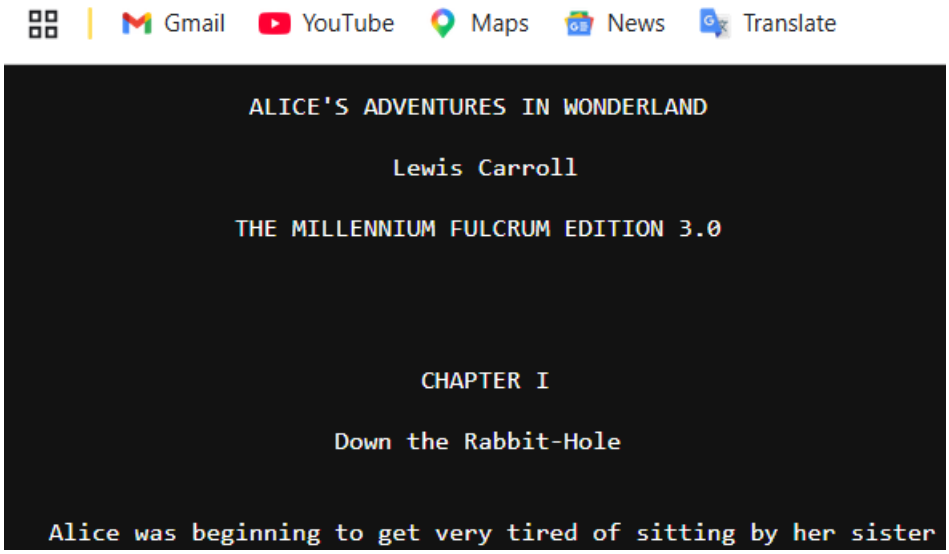
SURABAYA

6.2. Menangkap Tansfer TCP dalam Jumlah Besar dari Komputer Pribadi ke Remote Server

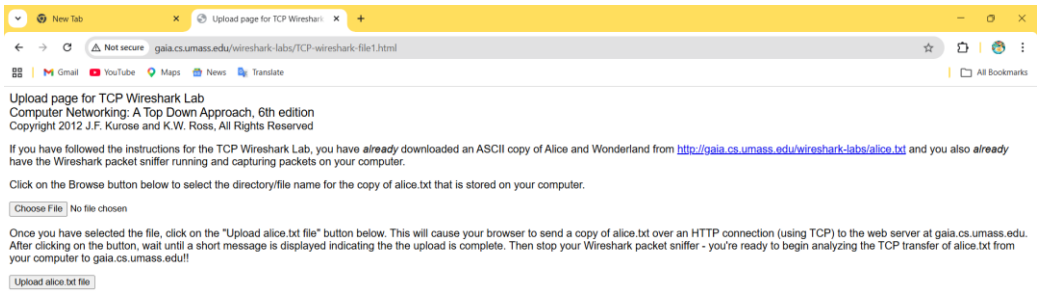
1) **Klik** http://gaia.cs.umass.edu/wireshark-labs/alice.txt



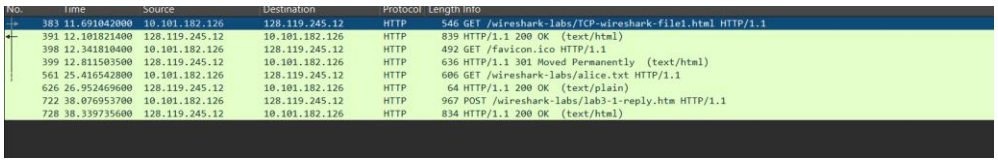
2) Save as file nya



3) **Klik upload dan Choose file**



4) **.**



No.	Time	Source	Destination	Protocol	Length	Info
383	11.691042000	10.101.182.126	128.119.245.12	HTTP	546	GET /wireshark-labs/TCP-wireshark-file1.html HTTP/1.1
391	12.101821400	128.119.245.12	10.101.182.126	HTTP	839	HTTP/1.1 200 OK (text/html)
398	12.341810400	10.101.182.126	128.119.245.12	HTTP	492	GET /favicon.ico HTTP/1.1
399	12.811503500	128.119.245.12	10.101.182.126	HTTP	636	HTTP/1.1 301 Moved Permanently (text/html)
561	25.416542800	10.101.182.126	128.119.245.12	HTTP	606	GET /wireshark-labs/alice.txt HTTP/1.1
626	26.952469600	128.119.245.12	10.101.182.126	HTTP	64	HTTP/1.1 200 OK (text/plain)
722	38.076953700	10.101.182.126	128.119.245.12	HTTP	967	POST /wireshark-labs/lab3-1-reply.htm HTTP/1.1
728	38.339735600	128.119.245.12	10.101.182.126	HTTP	834	HTTP/1.1 200 OK (text/html)

6.3. Tampilan Awal pada Captured Trace

1) **Berapa alamat IP dan nomor port TCP yang digunakan oleh komputer klien (sumber) untuk mentransfer file ke gaia.cs.umass.edu? Cara paling mudah menjawab pertanyaan ini adalah dengan memilih sebuah pesan HTTP dan meneliti detail paket TCP yang digunakan untuk membawa pesan HTTP tersebut.**

   Alamat IP Klien: 10.101.182.126

   Nomor Port TCP Klien: 55723

```
▾ Transmission Control Protocol, Src Port: 55723, Dst Port: 80, Seq: 1, Ack: 1, Len: 913
    Source Port: 55723
    Destination Port: 80
    [Stream index: 23]
    [Stream Packet Number: 4]
  ▸ [Conversation completeness: Complete, WITH_DATA (31)]
    [TCP Segment Len: 913]
    Sequence Number: 1    (relative sequence number)
    Sequence Number (raw): 798164039
    [Next Sequence Number: 914    (relative sequence number)]
    Acknowledgment Number: 1    (relative ack number)
    Acknowledgment number (raw): 1918500761
    0101 .... = Header Length: 20 bytes (5)
  ▸ Flags: 0x018 (PSH, ACK)
    Window: 259
    [Calculated window size: 66304]
    [Window size scaling factor: 256]
    Checksum: 0x4974 [unverified]
    [Checksum Status: Unverified]
    Urgent Pointer: 0
  ▸ [Timestamps]
```

No.	Time	Source	Destination	Protocol	Length	Info
704	36.878432800	10.101.182.126	128.119.245.12	TLSv1.3	134	Change Cipher Spec, Application Data
705	36.883905400	10.101.182.126	128.119.245.12	TCP	54	61809 → 443 [FIN, ACK] Seq=2781 Ack=1077 Win=65280 Len=0
706	37.182421100	128.119.245.12	10.101.182.126	TLSv1.3	373	Application Data
707	37.182421100	128.119.245.12	10.101.182.126	TLSv1.3	78	Application Data
708	37.182421100	128.119.245.12	10.101.182.126	TCP	54	443 → 61809 [FIN, ACK] Seq=1101 Ack=2782 Win=65536 Len=0
709	37.182564700	10.101.182.126	128.119.245.12	TCP	54	61809 → 443 [RST, ACK] Seq=2782 Ack=1101 Win=0 Len=0
710	37.233642500	10.101.182.126	128.119.245.12	TCP	54	65426 → 443 [ACK] Seq=2207 Ack=580 Win=65536 Len=0
721	38.076404600	10.101.182.126	128.119.245.12	TCP	66	57293 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
722	38.076953700	10.101.182.126	128.119.245.12	HTTP	967	POST /wireshark-labs/lab3-1-reply.htm HTTP/1.1
725	38.330388700	128.119.245.12	10.101.182.126	TCP	66	80 → 57293 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1412 SACK_PERM WS=128
726	38.330546900	10.101.182.126	128.119.245.12	TCP	54	57293 → 80 [ACK] Seq=1 Ack=1 Win=66304 Len=0
727	38.339735600	128.119.245.12	10.101.182.126	TCP	54	80 → 55723 [ACK] Seq=1 Ack=914 Win=63360 Len=0
728	38.339735600	128.119.245.12	10.101.182.126	HTTP	834	HTTP/1.1 200 OK (text/html)
729	38.381979600	10.101.182.126	128.119.245.12	TCP	54	55723 → 80 [ACK] Seq=914 Ack=781 Win=65536 Len=0
731	38.424795200	10.101.182.126	128.119.245.12	TLSv1.3	596	Application Data
734	38.584459000	10.101.182.126	172.217.194.188	TCP	55	61451 → 5228 [ACK] Seq=1 Ack=1 Win=254 Len=1
736	38.620122900	172.217.194.188	10.101.182.126	TCP	66	5228 → 61451 [ACK] Seq=1 Ack=2 Win=1048 Len=0 SLE=1 SRE=2
737	38.689561100	128.119.245.12	10.101.182.126	TLSv1.3	551	Application Data
738	38.692785000	10.101.182.126	128.119.245.12	TCP	54	65426 → 443 [FIN, ACK] Seq=2749 Ack=1077 Win=65280 Len=0
743	39.025784100	128.119.245.12	10.101.182.126	TLSv1.3	78	Application Data

2) **Apa alamat IP dari gaia.cs.umass.edu? Pada nomor port berapa ia mengirim dan menerima segmen TCP untuk koneksi ini?**

   Alamat IP Server: 128.119.245.12

   Nomor Port TCP Server: 80

```
▸ Frame 722: Packet, 967 bytes on wire (7736 bits), 967 bytes captured (7736 bits) on interface \Device\NP
▸ Ethernet II, Src: Intel_fa:27:b0 (48:45:20:fa:27:b0), Dst: 4a:58:76:94:b6:e3 (4a:58:76:94:b6:e3)
▸ Internet Protocol Version 4, Src: 10.101.182.126, Dst: 128.119.245.12
▸ Transmission Control Protocol, Src Port: 55723, Dst Port: 80, Seq: 1, Ack: 1, Len: 913
▸ Hypertext Transfer Protocol
▸ MIME Multipart Media Encapsulation, Type: multipart/form-data, Boundary: "----WebKitFormBoundarypPAu6WBZ
```

```
▾ Transmission Control Protocol, Src Port: 55723, Dst Port: 80, Seq: 1, Ack: 1, Len: 913
    Source Port: 55723
    Destination Port: 80
    [Stream index: 23]
    [Stream Packet Number: 4]
```

```
C:\Users\ASUS ROG STRIX>nslookup
Default Server:  UnKnown
Address:  10.101.182.209

> gaia.cs.umass.edu
Server:  UnKnown
Address:  10.101.182.209

Non-authoritative answer:
Name:    gaia.cs.umass.edu
Addresses:  64:ff9b::8077:f50c
          128.119.245.12

>
```

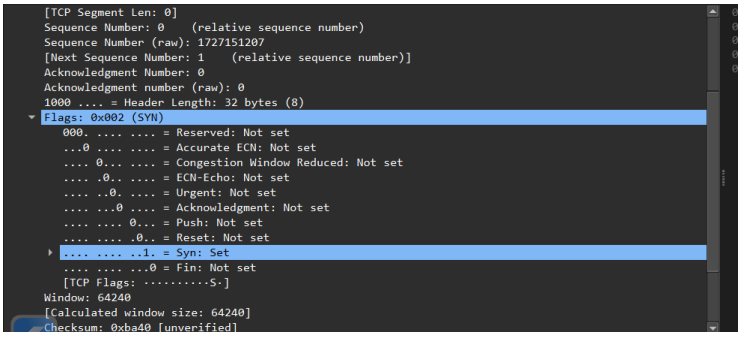
3) **Berapa alamat IP dan nomor port TCP yang digunakan oleh komputer klien Anda (sumber) untuk mentransfer ke gaia.cs.umass.edu?**

Alamat IP Klien (Saya): 10.101.182.126

Nomor Port TCP Klien (Saya): 55723

6.4. Dasar TCP

1) **Berapa nomor urut segmen TCP SYN yang digunakan untuk memulai sambungan TCP antara komputer klien dan gaia.cs.umass.edu? Apa yang dimiliki segmen tersebut sehingga teridentifikasi sebagai segmen SYN?**

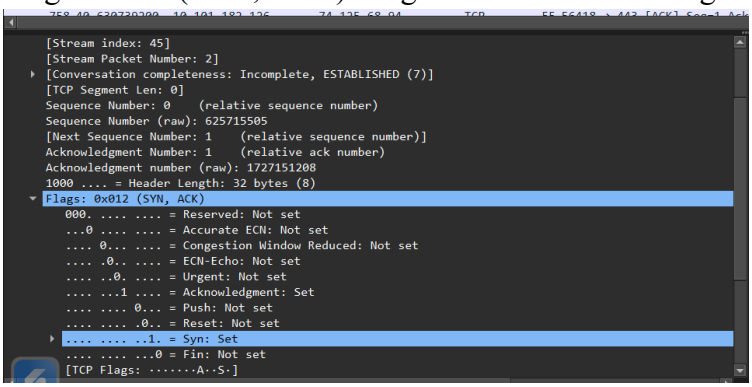


2) **Berapa nomor urut segmen SYNACK yang dikirim oleh gaia.cs.umass.edu ke komputer klien sebagai balasan dari SYN? Berapa nilai dari field Acknowledgement pada segmen SYNACK? Bagaimana gaia.cs.umass.edu menentukan nilai tersebut? Apa yang dimiliki oleh segmen sehingga teridentifikasi sebagai segmen SYNACK?**

Sequence Number: 0

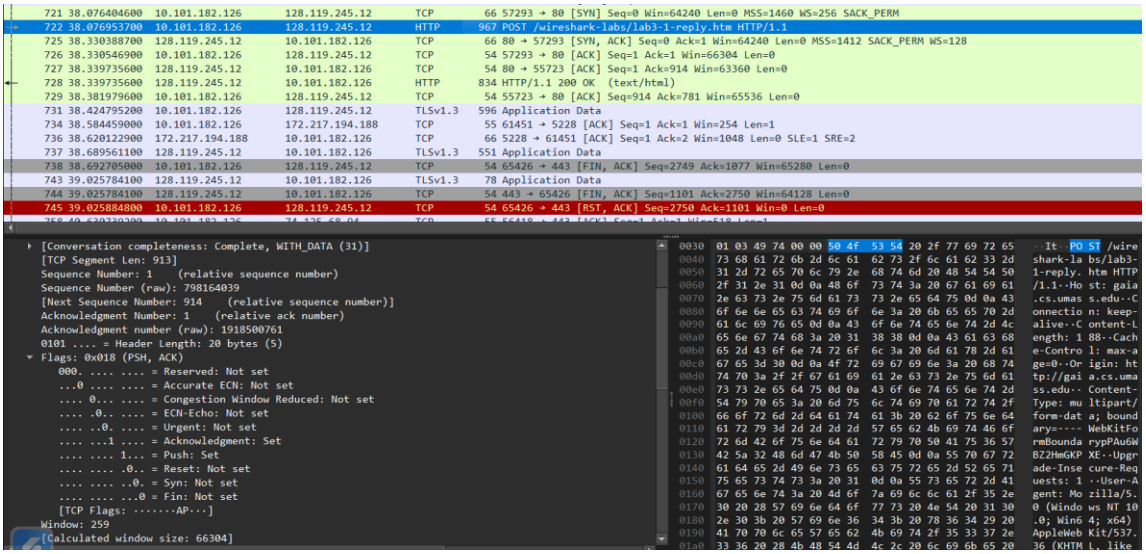
Acknowledgment Number: 1

Flags: 0x012 (SYN, ACK) dengan rincian Acknowledgment: Set dan Syn: Set



3) **Berapa nomor urut segmen TCP yang berisi perintah HTTP POST? Perhatikan bahwa untuk menemukan perintah POST, Anda harus menelusuri content field milik paket di bagian bawah jendela Wireshark, kemudian cari segmen yang berisi "POST" di bagian field DATAnya.**

Nomor Urut Segmen (Sequence Number): Nomor urut segmen TCP yang berisi perintah HTTP POST adalah 1.



4) Anggap segmen TCP yang berisi HTTP POST sebagai segmen pertama dalam koneksi TCP.

Berapa nomor urut dari enam segmen pertama dalam TCP (termasuk segmen yang berisi HTTP POST)? Pada jam berapa setiap segmen dikirim? Kapan ACK untuk setiap segmen diterima? Dengan adanya perbedaan antara kapan setiap segmen TCP dikirim dan kapan acknowledgement-nya diterima, berapakah nilai RTT untuk keenam segmen tersebut? Berapa nilai EstimatedRTT setelah penerimaan setiap ACK? (Catatan: Wireshark memiliki fitur yang memungkinkan Anda untuk memplot RTT untuk setiap segmen TCP yang dikirim. Pilih segmen TCP yang dikirim dari klien ke server gaia.cs.umass.edu pada jendela "daftar

paket yang ditangkap". Kemudian pilih: Statistics->TCP Stream Graph- >Round Trip Time Graph).

5) **Berapa panjang setiap enam segmen TCP pertama?**
segmen TCP pertama: 913 byte.

721	38.076404600	10.101.182.126	128.119.245.12	TCP	66	57293 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
722	38.076953700	10.101.182.126	128.119.245.12	TCP	967	55723 → 80 [PSH, ACK] Seq=1 Ack=1 Win=66304 Len=913
725	38.330388700	128.119.245.12	10.101.182.126	TCP	66	80 → 57293 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1412 SACK_PERM WS=128
726	38.330546900	10.101.182.126	128.119.245.12	TCP	54	57293 → 80 [ACK] Seq=1 Ack=1 Win=66304 Len=0
727	38.339735600	128.119.245.12	10.101.182.126	TCP	54	80 → 55723 [ACK] Seq=1 Ack=914 Win=63360 Len=0
728	38.339735600	128.119.245.12	10.101.182.126	TCP	834	80 → 55723 [PSH, ACK] Seq=1 Ack=914 Win=63360 Len=780
729	38.381979600	10.101.182.126	128.119.245.12	TCP	54	55723 → 80 [ACK] Seq=914 Ack=781 Win=65536 Len=0

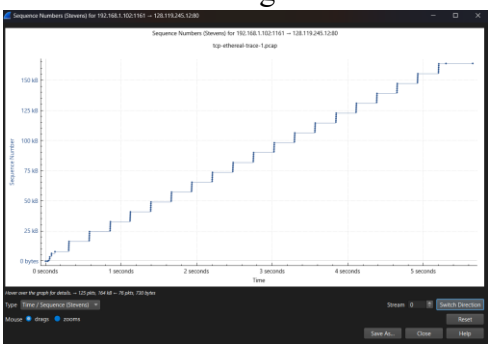
6) **Berapa jumlah minimum ruang buffer tersedia yang disarankan kepada penerima dan diterima untuk seluruh trace? Apakah kurangnya ruang buffer penerima pernah menghambat pengiriman?**
Paket 725: Win=64240
Paket 727: Win=63360
Paket 729: Win=65536

721	38.076404600	10.101.182.126	128.119.245.12	TCP	66	57293 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
722	38.076953700	10.101.182.126	128.119.245.12	TCP	967	55723 → 80 [PSH, ACK] Seq=1 Ack=1 Win=66304 Len=913
725	38.330388700	128.119.245.12	10.101.182.126	TCP	66	80 → 57293 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1412 SACK_PERM WS=128
726	38.330546900	10.101.182.126	128.119.245.12	TCP	54	57293 → 80 [ACK] Seq=1 Ack=1 Win=66304 Len=0
727	38.339735600	128.119.245.12	10.101.182.126	TCP	54	80 → 55723 [ACK] Seq=1 Ack=914 Win=63360 Len=0
728	38.339735600	128.119.245.12	10.101.182.126	TCP	834	80 → 55723 [PSH, ACK] Seq=1 Ack=914 Win=63360 Len=780
729	38.381979600	10.101.182.126	128.119.245.12	TCP	54	55723 → 80 [ACK] Seq=914 Ack=781 Win=65536 Len=0

7) **Apakah ada segmen yang ditransmisikan ulang dalam file trace? Apa yang anda periksa (di dalam file trace) untuk menjawab pertanyaan ini?**
Tidak, **tidak ada** segmen TCP yang ditransmisikan ulang dalam proses transfer data ini. cuma 1 segmen

6.5.Congestion Control pada TCP

1) **Gunakan alat plotting Time-Sequence-Graph (Stevens) untuk melihat grafik nomor urut berbanding waktu dari segmen yang dikirim oleh klien ke server gaia.cs.umass.edu. Dapatkah Anda mengidentifikasi di mana fase "slow start" TCP dimulai dan berakhir, dan pada bagian mana algoritma "congestion avoidance" mengambil alih? Berikan komentar tentang bagaimana data yang diukur berbeda dari perilaku ideal TCP yang telah kita pelajari.**
fase Slow Start teridentifikasi di awal transmisi dari detik 0,0 hingga sekitar 0,3 yang ditandai dengan kenaikan nomor urut secara eksponensial (melengkung), sebelum akhirnya beralih ke fase Congestion Avoidance hingga detik 5,5 yang polanya berubah menjadi linier (garis lurus miring). Berbeda dengan perilaku ideal TCP yang biasanya digambarkan sebagai kurva mulus, grafik realita ini membentuk pola anak tangga (staircase) di mana setiap garis vertikal menunjukkan pengiriman sekelompok paket secara sekaligus (*burst*) dan garis horizontal menunjukkan waktu tunggu *Round Trip Time* (RTT) untuk menerima konfirmasi (ACK) dari server sebelum mengirimkan data berikutnya.



No.	Time	Source	Destination	Protocol	Length	Info
25	0.400164	128.119.245.12	192.168.1.102	TCP	60	80 → 1161 [ACK] Seq=1 Ack=11933 Win=29200 Len=0
26	0.448613	128.119.245.12	192.168.1.102	TCP	60	80 → 1161 [ACK] Seq=1 Ack=13393 Win=32120 Len=0
27	0.500029	128.119.245.12	192.168.1.102	TCP	60	80 → 1161 [ACK] Seq=1 Ack=14853 Win=35040 Len=0
28	0.545852	128.119.245.12	192.168.1.102	TCP	60	80 → 1161 [ACK] Seq=1 Ack=16313 Win=37960 Len=0
29	0.576417	128.119.245.12	192.168.1.102	TCP	60	80 → 1161 [ACK] Seq=1 Ack=17285 Win=37960 Len=0
30	0.576671	192.168.1.102	128.119.245.12	TCP	1514	1161 → 80 [ACK] Seq=17285 Ack=1 Win=17520 Len=1460
31	0.577385	192.168.1.102	128.119.245.12	TCP	1514	1161 → 80 [ACK] Seq=18665 Ack=1 Win=17520 Len=1460
32	0.578329	192.168.1.102	128.119.245.12	TCP	1514	1161 → 80 [ACK] Seq=20125 Ack=1 Win=17520 Len=1460
33	0.579195	192.168.1.102	128.119.245.12	TCP	1514	1161 → 80 [ACK] Seq=21585 Ack=1 Win=17520 Len=1460
34	0.580149	192.168.1.102	128.119.245.12	TCP	1514	1161 → 80 [ACK] Seq=23045 Ack=1 Win=17520 Len=1460
35	0.581074	192.168.1.102	128.119.245.12	TCP	948	1161 → 80 [PSH, ACK] Seq=24505 Ack=1 Win=17520 Len=892
36	0.626496	128.119.245.12	192.168.1.102	TCP	60	80 → 1161 [ACK] Seq=1 Ack=18665 Win=40880 Len=0
37	0.672796	128.119.245.12	192.168.1.102	TCP	60	80 → 1161 [ACK] Seq=1 Ack=20125 Win=43800 Len=0
38	0.730684	128.119.245.12	192.168.1.102	TCP	60	80 → 1161 [ACK] Seq=1 Ack=21585 Win=46720 Len=0
39	0.772990	128.119.245.12	192.168.1.102	TCP	60	80 → 1161 [ACK] Seq=1 Ack=23045 Win=49640 Len=0
40	0.820622	128.119.245.12	192.168.1.102	TCP	60	80 → 1161 [ACK] Seq=1 Ack=24505 Win=52560 Len=0
41	0.853186	128.119.245.12	192.168.1.102	TCP	60	80 → 1161 [ACK] Seq=1 Ack=25397 Win=52560 Len=0
42	0.853405	192.168.1.102	128.119.245.12	TCP	1514	1161 → 80 [ACK] Seq=25397 Ack=1 Win=17520 Len=1460
43	0.854076	192.168.1.102	128.119.245.12	TCP	1514	1161 → 80 [ACK] Seq=26857 Ack=1 Win=17520 Len=1460
44	0.855036	192.168.1.102	128.119.245.12	TCP	1514	1161 → 80 [ACK] Seq=28317 Ack=1 Win=17520 Len=1460